LECTURE 01

NUMBER SYSTEM

INTRODUCTION TO CALCULUS

- Calculus is the mathematical tool used to analyze changes in physical quantities.
- Calculus is also mathematics of motion and change.
- Where there is motion of growth, where variable forces are at work producing acceleration, calculus is right mathematics to apply.

DIFFERENTIAL CALCULUS

Deals with the problem of finding:

- Rate of change
- Slope of Curve

Examples:

- Velocities and acceleration of moving bodies
- Firing angles that give cannons their maximum range
- The time where planets would be closest together or farthest apart

INTEGRAL CALCULUS

Deals with the problem of determining function from information about its rate of change.

Integral calculus enables us:

- To calculate lengths of curves
- To find areas of irregular region in plane
- To find the volumes and masses of arbitrary solids.
- To calculate the future location of a body from its present position and knowledge of the forces acting on it.

NUMBER SYSTEM

A number is a mathematical object used to count, measure, & label. It is the mathematical notation for representing numbers of a given set by using digits or other symbols in a consistent manner. It provides a unique representation of every number and represents the arithmetic and algebraic structure of the figures.

TYPES OF STANDARD NUMBERS

Natural Numbers

The set of positive integers that start at 1 and continue infinitely. The set of natural numbers is represented by the letter 'N' or the symbol $\mathbb{N}$.

$$N = \{1,2,3,4,5 \ldots.\}$$

Whole Numbers

In mathematics, whole numbers are positive integers, including zero, that do not have any decimal or fractional parts. The symbol for whole numbers is W

$$W = \{0, 1, 2, 3, 4, 5 \ldots\}$$

CONT…

Integers

Integers, also known as whole numbers or round numbers, are positive or negative numbers that don't have fractional or decimal parts. The symbol for integers is Z.

$$Z = \{\ldots, -1, -2, 0, 1, 2, \ldots\}$$

Rational Numbers

The set of rational numbers includes all the integers, each of which can be written as a quotient with the integer. A number that can be represented as the quotient p/q of two integers.

$$\mathrm{Q} = \{x: x = \frac{p}{q}, q \neq 0, p, q \epsilon Z\}$$

$$Q = \{3/2, 1/5, 3/4 \ldots\}$$

CONT…

Irrational Numbers

An irrational number is a real number that cannot be written as a fraction of two integers.

$$\mathrm{Q}' = \{x : x \neq \frac{p}{q}, q \neq 0, p, q \epsilon Z\}$$

$$Q' = \{\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi, \dots\}$$

Real Numbers

Real numbers can be positive or negative & include fractions, integers & irrational numbers. They can be used in arithmetic operations and represented on a number line.

$$R \;=\; \{23, 12, 6.99, 5/2, \pi, \sqrt{2}, 12.6\}$$

CONT…

Prime Numbers

A prime number is a whole number that is greater than 1 and can only be divided by itself and 1 without a remainder. For example, 19 is a prime number because it can only be divided by 1 and 19 these are

$$P \ = \{2,3,5,7,11 \dots\}$$

Complex Numbers

A complex number is a number that has both real and imaginary parts, and is written in the form

$$C = \{a + ib, a, b \in R\}$$

For example $C = \{2 + i0 = 2, 1 + 3i\}$

CONT…

Even Numbers

Numbers that can be divided into two equal parts, while odd numbers are numbers that cannot.

Even numbers: $0, \pm 2, \pm 4, \pm 6, \ldots \ldots ..$

Odd numbers: $\pm 1, \pm 3, \pm 5, \ldots \ldots \ldots$

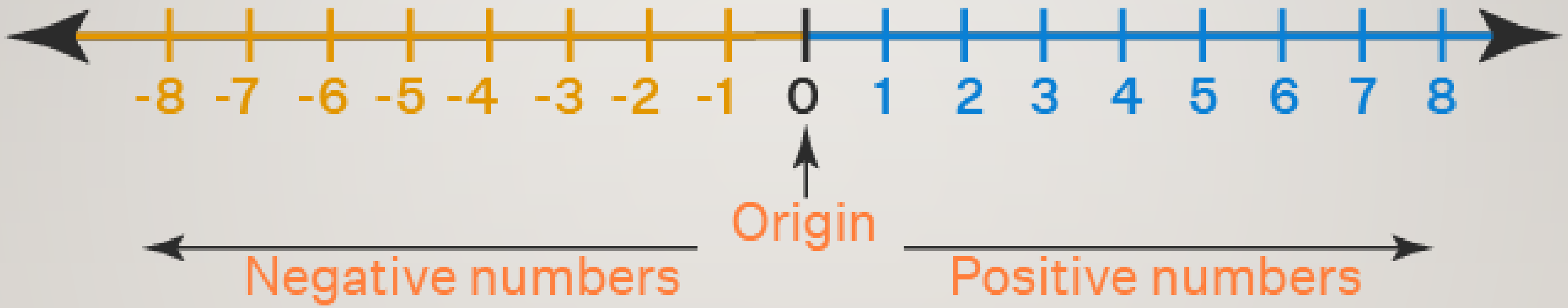
REMEMBER

- $N \;=\; \{1,2,3,4,5 \,\ldots .\}$
- $W \;=\; \{0, 1, 2, 3, 4, 5 \ldots\}$
- $Z \;=\; \{\ldots, -2, -1,0,1,2, \ldots\}$
- $\mathrm{Q} = \{x: x = \frac{p}{q}, q \neq 0, p, q\epsilon Z\}$
- $\mathrm{Q}' = \left\{x: x \neq \frac{p}{q}, q \neq 0, p, q\epsilon Z\right\}$
- $R = Q \cup Q'$
- $C = \{a + ib, a, b \in R\}$
- $N \subseteq W \subseteq Z \subseteq R \subseteq C$

NUMBER LINE

- A number line is a straight line with numbers placed at equal intervals along its length.
- It helps visualize numbers and their order.
- The center is usually zero, with positive numbers to the right and negative numbers to the left.

CONT…



-8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8
Origin
Negative numbers
Positive numbers

Symbols

$>$

greater than

$<$

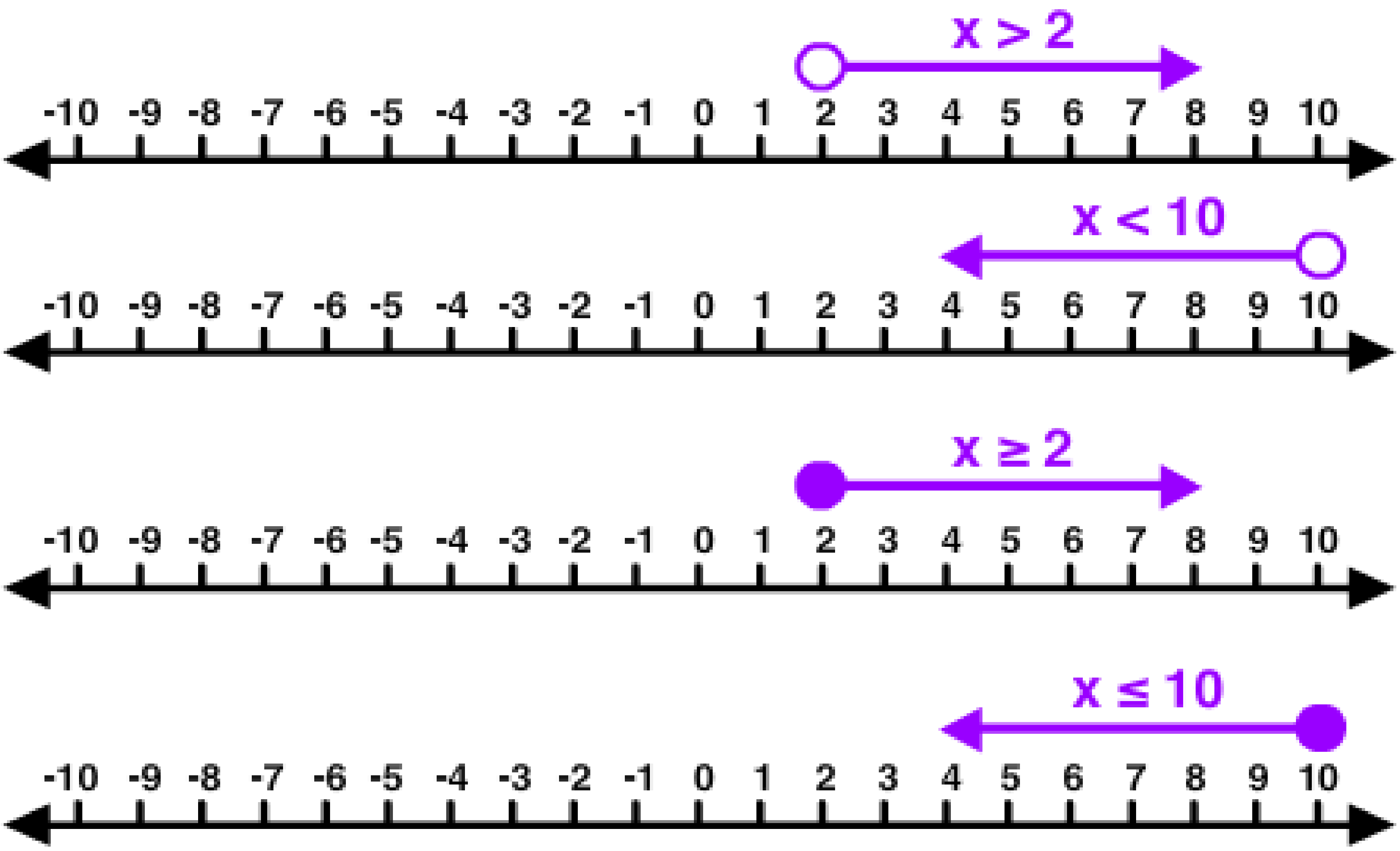
less than

$\geq$

greater than or equal to

$\leq$

less than or equal to



INTERVAL

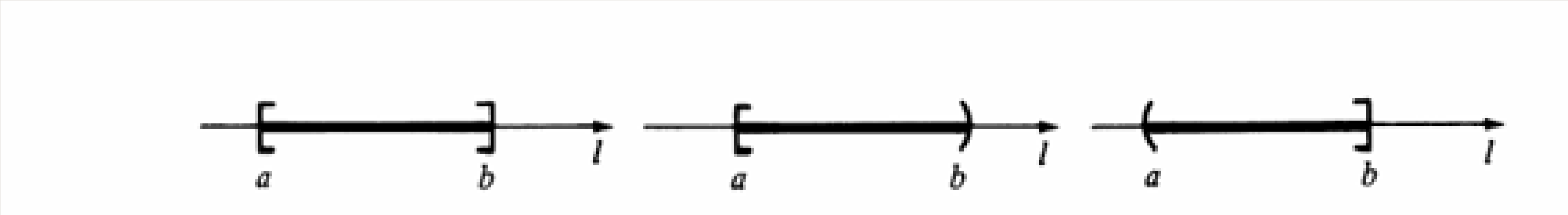
Interval is the set of real numbers in which every number that lies between any two numbers.

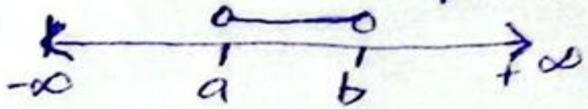
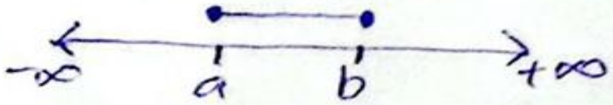
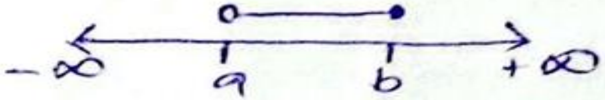
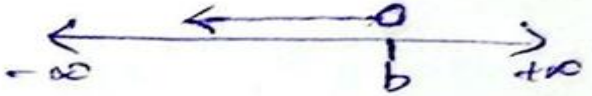
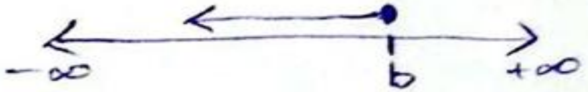
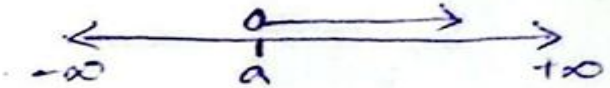
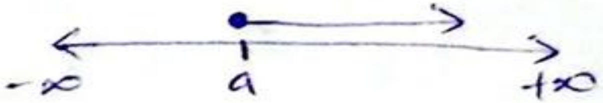
Two types of interval are:

- Open Interval---Between ( ) or (a, b)
- Closed Interval---from [ ] or [a, b]

CONT…

$$[a, b] = \{x: a \leq x \leq b\}$$
$$[a, b) = \{x: a \leq x < b\}$$
$$(a, b] = \{x: a < x \leq b\}$$



Interval	Set Notation	Geometry	Classification
(a, b)	$a < x < b$		open
$[a, b]$	$a \le x \le b$		closed
$(a, b]$	$a < x \le b$		semi open
$[a, b)$	$a \le x < b$		semi open
$(-\infty, b)$	$-\infty < x < b$		open
$(-\infty, b]$	$-\infty < x \le b$		semi open
$(a, +\infty)$	$x > a$		open
$[a, +\infty)$	$x \ge a$		semi open
$(-\infty, +\infty)$	$-\infty < x < +\infty$ or $x \in \mathbb{R}$		open